

LABORATORY OBSERVATION / RECORD

Course:	Full Stack Web Development
Course Code:	23CM4121
Experiment No.:	1
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TASK 1 — HTML AND HTML5

1. TITLE

Develop a College Information Portal Using HTML and HTML5 Features

2. AIM

To understand and implement the fundamental building blocks of HTML and HTML5 for creating a structured and interactive web page containing text, lists, tables, images, forms, semantic elements, multimedia, and HTML5 input types.

3. OBJECTIVE

The objectives of this task are:

- To understand the basic structure of an HTML5 document.
- To create web pages using headings and paragraphs.
- To apply different text-formatting tags.
- To create ordered and unordered lists.
- To add hyperlinks to external websites.
- To create and display student details using tables.
- To insert images with alternate text.
- To design a student registration form.
- To use HTML5 semantic elements.
- To embed audio and video files.
- To implement HTML5 input types such as email, date, number, color, and range.
- To understand the role of HTML in structuring web content.

4. THEORY

HTML (HyperText Markup Language) is the standard markup language used to create and structure web pages. HTML uses elements and tags to define different parts of a web page such as headings, paragraphs, lists, tables, forms, images, and links.

HTML5 is the modern version of HTML that provides additional features such as semantic elements, multimedia support, canvas, and new form input types.

The basic structure of an HTML5 document consists of:

- `<!DOCTYPE html>` — Defines the document as an HTML5 document.
- `<html>` — Root element of the HTML document.
- `<head>` — Contains metadata and the page title.
- `<body>` — Contains the visible content of the web page.

Important concepts used in this task:

Headings and Paragraphs: `<h1>` to `<h6>` are used for headings, while `<p>` is used for paragraphs.

Text Formatting: `<b>` makes text bold, `<i>` makes text italic, `<u>` underlines text, `<sup>` displays superscript text, and `<sub>` displays subscript text.

Lists: `<ul>` creates an unordered list, while `<ol>` creates an ordered list. Individual items are created using `<li>`.

Hyperlinks: The `<a>` tag is used to create links to other web pages or websites.

Tables: `<table>`, `<tr>`, `<th>`, and `<td>` are used to create and organize tabular information.

Images: The `<img>` tag is used to display images. The `alt` attribute provides alternative text if the image cannot be displayed.

Forms: HTML forms collect user information using controls such as text boxes, radio buttons, checkboxes, dropdowns, textareas, and buttons.

Semantic Elements: HTML5 provides meaningful elements such as `<header>`, `<nav>`, `<section>`, `<article>`, and `<footer>` to describe the structure of a web page.

Multimedia: The `<audio>` and `<video>` elements allow audio and video files to be embedded directly into web pages.

HTML5 Input Types: HTML5 provides specialized input types such as email, date, number, color, and range, which provide better input handling and browser-level validation.

5. ALGORITHM / PROCEDURE

1. Create a new HTML file.
2. Define the document as an HTML5 document using `<!DOCTYPE html>`.
3. Create the basic `<html>`, `<head>`, and `<body>` structure and set the page title to College Information Portal.
4. Wrap the page content in a `<center>` tag so the portal is centrally aligned.
5. Create a `<header>` section with the portal heading and a welcome message.
6. Create a `<nav>` bar with links to the About, Students, Registration, and Media sections using anchor (#) links.
7. Create a section with `id="about"` containing an `<article>` with a welcome heading and description.
8. Apply bold, italic, and underline formatting to highlight accreditation and facility details.
9. Create an ordered list of the college Departments.
10. Create an unordered list of the college Facilities.
11. Add hyperlinks to the college website and Google, opening in a new tab using `target="_blank"`.
12. Create a section with `id="students"` containing a Student Details table with Roll No, Name, Branch, and Email columns.
13. Insert the college logo and college building images using the `<img>` tag with appropriate alt text.
14. Create a section with `id="reg"` containing a `<form>` wrapped in a `<fieldset>` with a `<legend>` titled Student Registration Form.
15. Add Name, Email, and Password input controls to the form.
16. Add Gender radio buttons and Hobbies checkboxes to the form.
17. Add a Branch dropdown using `<select>` and a Comments `<textarea>`.
18. Add HTML5 input types — color, date, number, and range — for Favorite Color, Date of Birth, Mobile Number, and Satisfaction Level.

19. Add Submit and Reset buttons to the form.
20. Create a section with id="media" containing an <audio> element and a <video> element with playback controls.
21. Add a <footer> with the copyright text.
22. Save the HTML file.
23. Open the file in a web browser.
24. Verify all headings, lists, links, table, images, navigation, form controls, audio, and video.
25. Record the output using screenshots.

6. PROGRAM

index.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>College Information Portal</title>
</head>
<body>
  <center>
    <!--Header Section-->
    <header>
      <h1>College Information Portal</h1>
      <p>Welcome to Anits College Portal</p>
    </header>
    <hr>
    <!--Navigation Bar-->
    <nav>
      <a href="#about">About</a> |
      <a href="#students">Students</a> |
      <a href="#reg">Registration</a> |
      <a href="#media">Media</a>
    </nav>
    <hr>
    <!--Part A-->
    <section id="about">
      <article>
        <h1>Welcome to Anits</h1><br>
        <p>Building future engineers with innovation and excellence.</p><br>
        <p>
          <b>Our institute provides quality education with experienced
faculty.</b><br>
          <i>Recognized by AICTE and affiliated with a reputed university. NAAC
"A+" Honoured</i><br>
          <u>Modern laboratories, smart classrooms, and excellent
placements.</u><br>
        </p>
        <h1>Departments</h1>
        <ol>
          <li>Computer Science</li>
          <li>Artificial Intelligence</li>
          <li>Data Science</li>
        </ol>
      </article>
    </section>
  </center>
</body>
</html>
```

```

        <li>Information Technology</li>
        <li>Electronics & Communication</li>
        <li>Electrical Engineering</li>
        <li>Mechanical Engineering</li>
        <li>Civil Engineering</li>
    </ol>
    <h1>Facilities</h1>
    <ul>
        <li>Central Library</li>
        <li>Computer Labs</li>
        <li>Sports Complex</li>
        <li>Hostel Facility</li>
        <li>Cafeteria</li>
    </ul>
    <h1>Useful Links</h1>
    <p>College Website</p>
    <a href="https://anits.org/" target="_blank">Anits Website</a>
    <p>Google Search</p>
    <a href="https://www.google.com" target="_blank">Google</a>

    </article>
</section>
<hr>
<!--Part B-->
<section id="students">
    <article>
        <h1>Student Details</h1>
        <table border="5" cellpadding="10">
            <tr>
                <th>Roll No</th>
                <th>Name</th>
                <th>Branch</th>
                <th>Email</th>
            </tr>
            <tr>
                <td>201</td>
                <td>Poojith</td>
                <td>CSE</td>
                <td>poojith4521@gmail.com</td>
            </tr>
            <tr>
                <td>091</td>
                <td>Suhas</td>
                <td>AI & ML</td>
                <td>suhas45@gmail.com</td>
            </tr>
            <tr>
                <td>203</td>
                <td>Sasank</td>
                <td>IT</td>
                <td>ksasank12@gmail.com</td>
            </tr>
        </table>
    </article>
</section>
<hr>

```

```

<section>
    
    
</section>
<hr>
<!--Part C-->
<section id="reg">
    <article>
        <form>
            <fieldset>
                <legend>
                    <h1>Student Registration Form</h1>
                </legend>
                <p>
                    <label for="name">Name</label>
                    <input type="text" id="name">
                    <br><br>
                    <label for="email">Email</label>
                    <input type="email" id="email">
                    <br><br>
                    <label for="password">Password</label>
                    <input type="password" id="password">
                    <br><br>
                    <label>Gender</label>
                    <input type="radio" name="gender">Male
                    <input type="radio" name="gender">Female
                    <input type="radio" name="gender">Other
                    <br><br>
                    <label>Hobbies</label>
                    <input type="checkbox">Reading
                    <input type="checkbox">Music
                    <input type="checkbox">Sports
                    <br><br>
                    <label for="branch">Branch</label>
                    <select id="branch">
                        <option>CSE</option>
                        <option>AI & ML</option>
                        <option>Data Science</option>
                        <option>IT</option>
                        <option>ECE</option>
                        <option>EEE</option>
                    </select>
                    <br><br>
                    <label for="comments">Comments</label>
                    <textarea id="comments" rows="6" cols="45" placeholder="Enter
your comments"></textarea>
                    <br><br>
                    <label for="color">Favorite Color</label>
                    <input type="color" id="color">
                    <br><br>
                    <label for="date">Date of Birth</label>
                    <input type="date" id="date">
                    <br><br>
                    <label for="number">Mobile Number</label>
                    <input type="number" id="number">
                    <br><br>

```

```

        <label for="range">Satisfaction Level</label>
        <input type="range" id="range">
        <br><br>
        <button type="submit">Submit</button>
        <button type="reset">Reset</button>
    </p>
</fieldset>
</form>
</article>
</section>
<hr>
<!--Part D-->
<section id="media">
    <article>
        <h1>College Audio File</h1>
        <audio controls>
            <source src="college_audio.mp3" type="audio/mpeg">
        </audio>
        <h1>College Video</h1>
        <video width="220" controls>
            <source src="college_video.mp4" type="video/mp4">
        </video>
    </article>
</section>
<hr>
<footer>
    <p>&copy; 2026 Anits College Info Portal. All Rights Reserved.</p>
</footer>
</center>
</body>
</html>

```

7. CODE EXPLANATION

Code / Tag	Explanation
<!DOCTYPE html>	Declares the document as HTML5.
<html>	Defines the root element of the web page.
<head>	Contains metadata and page information.
<title>	Specifies the title displayed in the browser tab.
<center>	Centrally aligns the page content.
<header>	Defines the header section of the page.
<nav>	Defines the navigation bar with links to page sections.
<section id="...">	Groups related content and provides an anchor target for navigation links.
<article>	Defines independent, self-contained content within a section.
<h1>	Defines the main heading used for each block of content.
<p>	Defines a paragraph.

	Displays text in bold.
<i>	Displays text in italic.
<u>	Underlines text.
	Creates an ordered list, used here for Departments.
	Creates an unordered list, used here for Facilities.
	Defines an individual list item.
<a>	Creates a hyperlink.
target="_blank"	Opens the linked page in a new browser tab.
<table>, <tr>, <th>, <td>	Create and organize the Student Details table.
, alt	Displays the college logo and building images with alternate text.
<form>	Creates the student registration form.
<fieldset>, <legend>	Groups the form controls and provides a titled border/caption.
<input>	Creates different types of input controls (text, email, password, radio, checkbox, color, date, number, range).
<select>	Creates the Branch dropdown list.
<textarea>	Creates the multi-line Comments input area.
<button type="submit"/"reset">	Submits or resets the registration form.
<audio>	Embeds the college audio file.
<video>	Embeds the college video file.
<footer>	Defines the footer section with copyright text.

8. TEST CASES AND EXECUTION DATA

Test Case	Test / Input	Expected Result	Actual Result	Status
TC-01	Open index.html	College Information Portal should be displayed	College Information Portal displayed	Pass
TC-02	Click a navigation bar link	Page should jump to the corresponding section (About/Students/Registration/Media)	Navigated to the correct section	Pass
TC-03	Check text formatting	Bold, italic and underline should appear correctly	Formatting displayed correctly	Pass
TC-04	Check Departments list	All 8 departments should be listed in order	Ordered list displayed correctly	Pass
TC-05	Check Facilities list	All 5 facilities should be listed	Unordered list displayed correctly	Pass
TC-06	Click College Website link	Anits website should open in a	Link opened successfully	Pass

		new tab		
TC-07	Click Google link	Google should open in a new tab	Link opened successfully	Pass
TC-08	Check Student Details table	3 student records should be displayed in tabular form	Table displayed correctly	Pass
TC-09	Check logo and building images	Images should be displayed with alternate text	Images displayed correctly	Pass
TC-10	Enter details in the registration form	User should be able to enter name, email, password, gender, hobbies, branch, comments, color, date, number and range	All form controls worked correctly	Pass
TC-11	Click Submit / Reset	Form should submit or reset correctly	Buttons worked correctly	Pass
TC-12	Check audio and video	Audio and video controls should be displayed in the Media section	Multimedia displayed successfully	Pass
TC-13	Check footer	Footer with copyright text should be displayed	Footer displayed correctly	Pass

9. OUTPUT

Output 1: College Information Portal

8/4/26, 10:21 AM

College Information Portal

College Information Portal

Welcome to Anits College Portal

[About](#) | [Students](#) | [Registration](#) | [Media](#)

Welcome to Anits

Building future engineers with innovation and excellence.

Our institute provides quality education with experienced faculty.

Recognized by AICTE and affiliated with a reputed university. NAAC "A+" Honoured
Modern laboratories, smart classrooms, and excellent placements.

Departments

1. Computer Science
2. Artificial Intelligence
3. Data Science
4. Information Technology
5. Electronics & Communication
6. Electrical Engineering
7. Mechanical Engineering
8. Civil Engineering

Facilities

- Central Library
- Computer Labs
- Sports Complex
- Hostel Facility
- Cafeteria

Useful Links

College Website

[Anits Website](#)

Google Search

[Google](#)

Figure 1(a): About, Departments, Facilities, Useful Links, and Student Details table.

Student Details

Roll No	Name	Branch	Email
201	Poojith	CSE	poojith4521@gmail.com
091	Suhas	AI & ML	suhas45@gmail.com
203	Sasank	IT	ksasank12@gmail.com

 College Logo  College Building

Student Registration Form

Name

Email

Password

Gender ☐ Male ☐ Female ☐ Other

Hobbies ☐ Reading ☐ Music ☐ Sports

Branch

Enter your comments

Comments

Favorite Color

Date of Birth

Mobile Number

Satisfaction Level

College Audio File

Figure 1(b): College logo, college building image, and the Student Registration Form.

0:00 / 0:00

College Video



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Figure 1(c): College Audio File, College Video, and footer.

10. INTERVIEW QUESTIONS

Q1. What is the purpose of semantic HTML5 elements?

Answer:

Semantic HTML5 elements provide meaningful information about the structure and purpose of different sections of a web page. Elements such as <header>, <nav>, <section>, <article>, and <footer> make the HTML code easier to understand, maintain, and interpret by browsers, search engines, and assistive technologies.

Q2. Compare Ordered and Unordered Lists.

Answer:

An ordered list () displays items in a specific sequence, usually using numbers or letters. An unordered list () displays items without indicating a particular order, usually using bullet points.

Q3. What happens if an image path is incorrect?

Answer:

If the image path is incorrect, the browser cannot find and display the image. Instead, the image will appear as broken or missing. The alternative text specified using the alt attribute may be displayed.

Q4. Differentiate Radio Buttons and Checkboxes.

Answer:

Radio buttons are generally used when the user must select one option from a group of options. Checkboxes are used when the user can select multiple independent options.

Q5. Explain the advantages of HTML5 input types over traditional text boxes.

Answer:

HTML5 input types provide specialized controls and built-in validation. For example, email validates email-format input, date provides a date selector, number supports numeric values, color provides a color picker, and range provides a slider. They improve usability and reduce the need for additional validation code.

11. RESULT

Thus, a College Information Portal was successfully developed using HTML and HTML5. The page implemented semantic sectioning (header, nav, section, article, footer) with anchor-based navigation, text formatting, ordered and unordered lists for Departments and Facilities, hyperlinks, a Student Details table, images with alternate text, and a fieldset-based Student Registration Form combining standard controls with HTML5 input types (email, password, color, date, number, and range). Audio and video were embedded successfully, and all elements were verified in the browser.